

Text Classification on 20-Newsgroups

From bag-of-words to large language models

Part 1

Simple NN

Part 2

BERT fine-tune

Part 3

MLM + fine-tune

Part 4

Llama-3 prompting

The problem & the dataset

The task

Given a Usenet post, assign it to one of 20 newsgroup categories.

Why it's interesting

- 20-way classification with lexically overlapping pairs
e.g. *ibm.pc.hardware* vs. *mac.hardware*
- Headers, footers, quoted text removed — model can't cheat
- Standard sklearn train/test split available

20-Newsgroups at a glance

~11,300 training documents

~7,500 test documents

20 balanced classes

Category families: *computing · recreation · science · politics · religion · for-sale*

A four-part progression

Each part addresses a specific limitation of the previous one.

PART 1

Simple NN + word reps

TF-IDF · Word2Vec · GloVe

Same 2-layer net, swap only the input encoding. Baseline understanding of where the signal lives.

PART 2

Fine-tune BERT

Full · Linear probe · Partial freeze

Replace bag-of-words with contextual [CLS] embeddings. Compare three training regimes.

PART 3

Domain-adaptive MLM

MLM on newsgroups → fine-tune

Continue BERT's masked-language modelling on newsgroup text before classification.

PART 4

Llama-3 prompting

Zero-shot · Few-shot

No training at all. Can a generative LLM classify with just instructions and 0–20 examples?

Same network, four input encodings

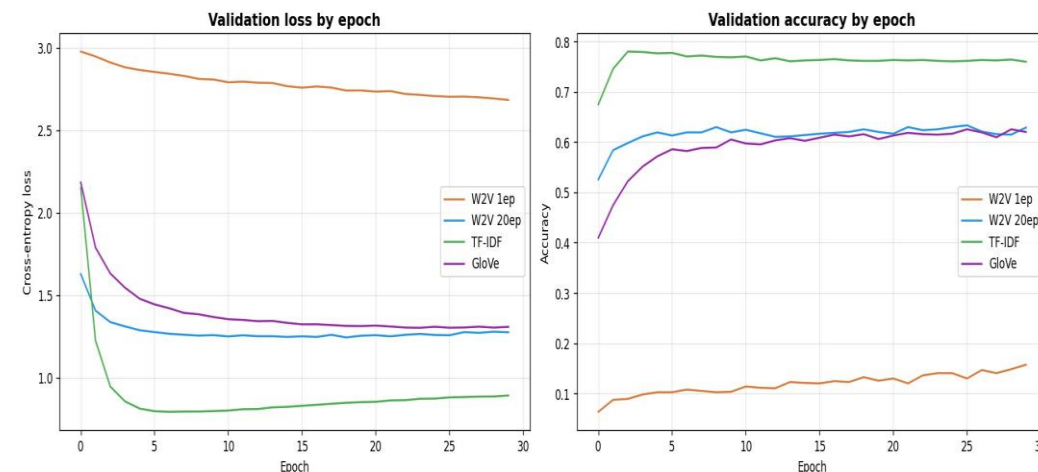
The control: a 2-layer MLP

Architecture and hyper-parameters are held constant across all four runs.
Any performance difference is attributable to the input representation alone.

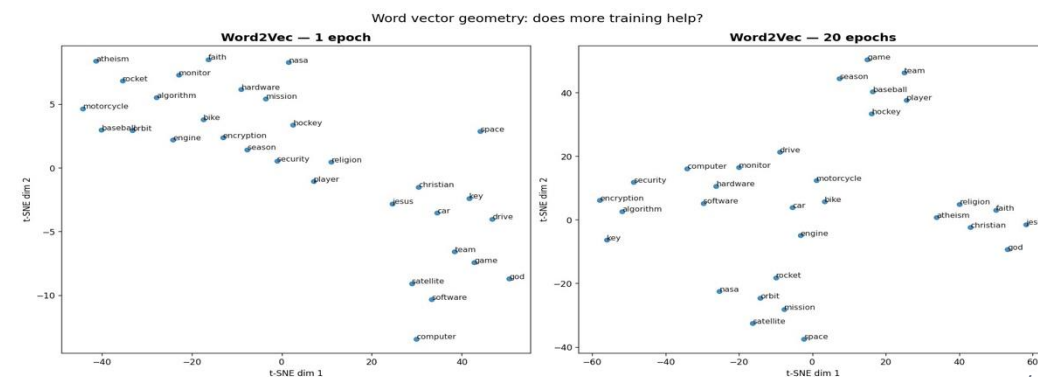
Four representations compared

Word2Vec (1 epoch)	Under-trained skip-gram baseline
Word2Vec (20 epochs)	Same setup, more passes — does training help?
TF-IDF (20k features)	Classic bag-of-words, term frequency $\times$ IDF
GloVe (pre-trained)	Wikipedia-trained 100-dim vectors, mean-pooled

Validation accuracy per epoch



Word2Vec embedding geometry (t-SNE)



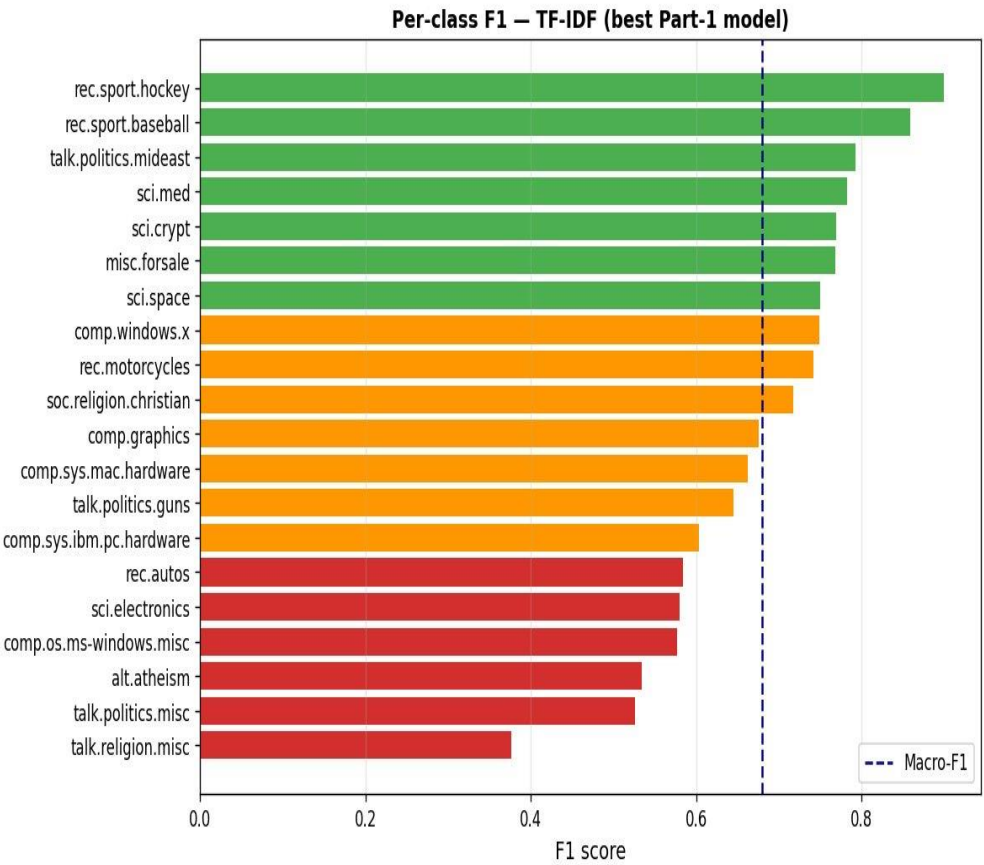
Results — TF-IDF wins, and that's instructive

Representation	Trainable params	Test accuracy	Macro-F1
Word2Vec — 1 epoch	30,996	14.4%	0.107
Word2Vec — 20 epochs	30,996	58.8%	0.571
GloVe — pre-trained	30,996	57.2%	0.552
TF-IDF (20k)	5,125,396	68.7%	0.680

Why does TF-IDF beat learned embeddings?

The signal is heavily lexical: *"crypt"* ≈ sci.crypt, *"atheism"* ≈ alt.atheism. TF-IDF keeps each rare keyword as its own dimension — no information loss. Word2Vec/GloVe mean-pool all word vectors, diluting strong signals across hundreds of background tokens.

Per-class F1 — best model



Fine-tuning BERT — three training regimes

How much of BERT's body do we need to update? Three configurations isolate the answer.

Run A Full fine-tune

109.5M

trainable parameters

All 110M BERT params + 15k head — most powerful, most expensive.

Run B Linear probe

15k

trainable parameters

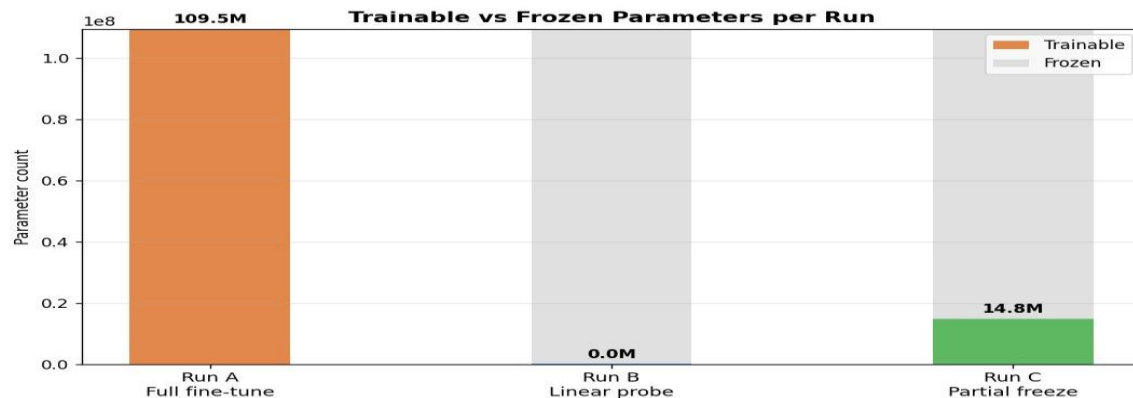
Freeze every BERT weight, train only the head. Tests raw pre-trained quality.

Run C Partial freeze

14.8M

trainable parameters

Freeze layers 0–9, train layers 10–11 + head. The practical sweet spot.



Why three runs?

The gap between A and B isolates the value of fine-tuning. Run C tests whether you can recover most of A's accuracy at a fraction of the compute.

Results — partial freeze recovers full fine-tune

Test macro-F1

Part 1 — TF-IDF baseline

0.6798

Run B — Linear probe

0.3337

Run C — Partial freeze

0.6880

Run A — Full fine-tune

0.7093

Confusion matrix — Run A (full fine-tune)

BERT full fine-tune — normalised confusion matrix (test set)

alt.atheism	0.51	0.01	0.00	0.00	0.00	0.01	0.00	0.00	0.03	0.01	0.01	0.01	0.00	0.01	0.03	0.19	0.06	0.02	0.04	0.06
comp.graphics	0.01	0.74	0.05	0.02	0.03	0.07	0.01	0.00	0.00	0.00	0.00	0.03	0.01	0.01	0.02	0.00	0.00	0.00	0.00	0.00
comp.os.ms-windows.misc	0.02	0.06	0.70	0.07	0.04	0.05	0.00	0.01	0.01	0.00	0.00	0.01	0.00	0.00	0.02	0.00	0.00	0.00	0.01	0.00
comp.sys.ibm.pc.hardware	0.00	0.03	0.13	0.67	0.05	0.01	0.02	0.00	0.01	0.00	0.00	0.02	0.06	0.00	0.00	0.00	0.00	0.00	0.00	0.00
comp.sys.mac.hardware	0.00	0.02	0.03	0.12	0.72	0.00	0.02	0.01	0.01	0.00	0.00	0.01	0.05	0.00	0.01	0.00	0.00	0.00	0.00	0.00
comp.windows.x	0.00	0.06	0.09	0.01	0.00	0.80	0.01	0.00	0.01	0.00	0.00	0.01	0.00	0.01	0.00	0.00	0.01	0.00	0.00	0.00
misc.forsale	0.00	0.01	0.01	0.05	0.02	0.00	0.86	0.01	0.01	0.00	0.01	0.01	0.01	0.01	0.00	0.01	0.00	0.00	0.00	0.00
rec.autos	0.01	0.00	0.01	0.00	0.01	0.00	0.02	0.77	0.09	0.00	0.01	0.00	0.05	0.00	0.01	0.00	0.02	0.01	0.01	0.00
rec.motorcycles	0.01	0.01	0.01	0.00	0.01	0.00	0.01	0.08	0.76	0.01	0.01	0.01	0.02	0.00	0.02	0.01	0.02	0.00	0.03	0.00
rec.sport.baseball	0.01	0.01	0.00	0.00	0.00	0.01	0.01	0.00	0.01	0.87	0.05	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.01	0.00
rec.sport.hockey	0.01	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.04	0.91	0.01	0.00	0.00	0.00	0.00	0.01	0.00	0.01	0.00
sci.crypt	0.02	0.00	0.01	0.01	0.01	0.00	0.01	0.00	0.01	0.01	0.00	0.74	0.05	0.01	0.01	0.00	0.05	0.01	0.04	0.01
sci.electronics	0.00	0.02	0.02	0.09	0.03	0.00	0.04	0.02	0.02	0.01	0.01	0.04	0.62	0.04	0.02	0.00	0.01	0.00	0.01	0.00
sci.med	0.01	0.02	0.00	0.00	0.01	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.02	0.87	0.00	0.01	0.01	0.01	0.02	0.01
sci.space	0.02	0.03	0.00	0.00	0.00	0.00	0.01	0.00	0.01	0.00	0.00	0.01	0.02	0.03	0.81	0.01	0.01	0.00	0.04	0.00
soc.religion.christian	0.08	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.83	0.01	0.00	0.01	0.07
talk.politics.guns	0.03	0.00	0.00	0.00	0.00	0.00	0.01	0.02	0.01	0.00	0.00	0.01	0.00	0.02	0.02	0.00	0.70	0.02	0.11	0.04
talk.politics.mideast	0.08	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.01	0.02	0.00	0.00	0.01	0.00	0.03	0.77	0.06	0.00
talk.politics.misc	0.06	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.02	0.01	0.00	0.01	0.00	0.03	0.03	0.00	0.29	0.03	0.49	0.02
talk.religion.misc	0.25	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.00	0.00	0.00	0.01	0.04	0.39	0.12	0.00	0.03	0.11

True label

Predicted label

Linear probe collapses

Frozen BERT alone scores 0.33 — its raw [CLS] representation is not class-discriminative for 20-way newsgroup classification.

Partial freeze ≈ full FT

Run C reaches 0.688 with only 14% of trainable params. Lower BERT layers don't need updating; upper layers do.

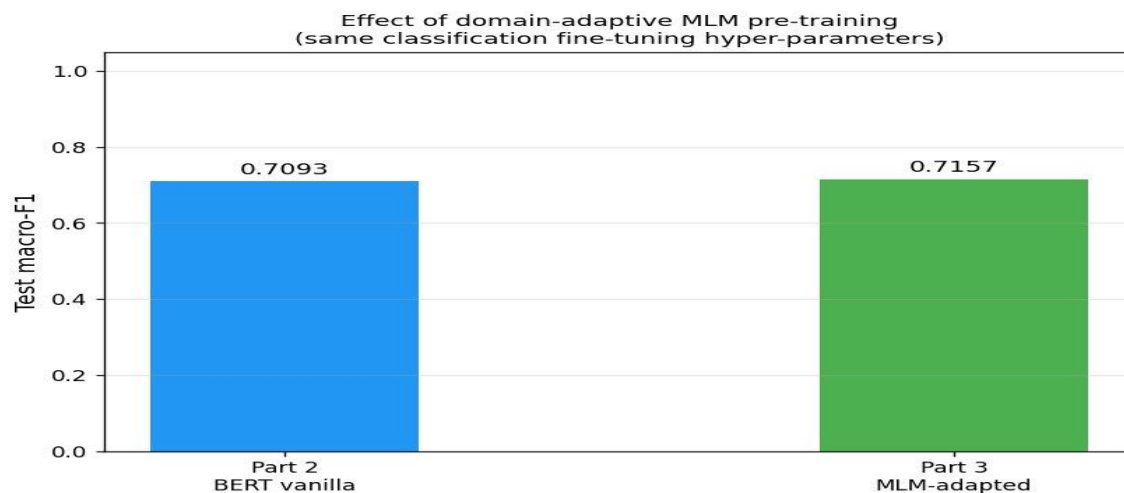
Domain-adaptive MLM pre-training

Stage 1 — continue BERT's masked language modelling on newsgroup text (no labels used)

Stage 2 — fine-tune the adapted checkpoint with identical hyper-parameters to Part 2A



Effect on test macro-F1



+ 0.6 pp macro-F1

0.7093 → 0.7157

Small but consistent. MLM nudges BERT toward in-domain terms (*nasa, crypt, dos, sgi*) and informal Usenet syntax — but bert-base-uncased was already trained on 3.3B tokens. The marginal value of 5M more is naturally limited.

Llama-3 — prompting instead of training

Can a generative LLM classify without any gradient updates?

Setup

Model: Llama-3-8B-Instruct via Groq API

Eval set: stratified sample, 20 docs × 20 classes = 400 posts

Decoding: temperature = 0 for reproducibility

Few-shot pool: drawn only from training split (no test leak)

Prompt engineering applied

- Role assignment
- Explicit output-format constraint (one line, exact label)
- Triple-quote delimiters around the post (injection defence)
- Robust 3-tier label parser (exact → substring → token)

Two regimes

Zero-shot

Prompt: task description + list of 20 class names + the post.
No labelled training data needed at all.

Few-shot (k = 1 per class)

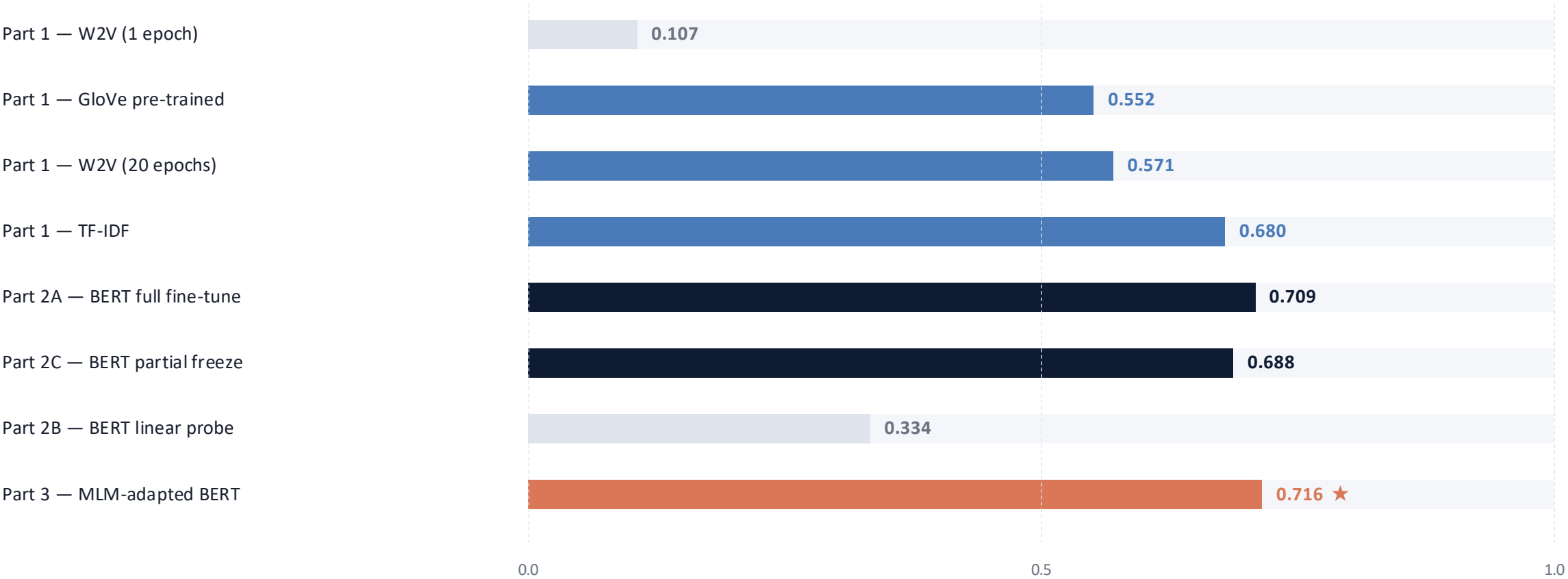
Same prompt + one labelled example per class (20 demos total).
Two effects: format anchoring + lexical priming.

Observed pattern

Zero-shot already classifies high-vocabulary topics (sci.space, rec.sport.hockey) well. Few-shot mainly helps the lexically confusable pairs.

Putting it all together

Test macro-F1 across all four parts



Best result: MLM-adapted BERT at 0.716 macro-F1 · + 3.6 pp over TF-IDF baseline · + 0.6 pp over vanilla BERT fine-tune

Takeaways & future work

What we learned

1 Representation > architecture (sometimes)

On a small lexically-rich corpus, sparse TF-IDF can hold its own against dense embeddings — mean-pooling is the bottleneck, not the embeddings.

2 Most of BERT doesn't need updating

Partial freeze of the bottom 10 layers recovers ~97% of full-fine-tune performance with 14% of the trainable params.

3 DAPT gains are real but modest here

Continuing MLM on newsgroup text adds 0.6 pp — meaningful and free, but small when the domain gap is small to begin with.

FUTURE WORK

Where to push next

Longer MLM stage

Train MLM for more epochs and on a larger in-domain corpus (Reddit, mailing lists) to amplify DAPT gain.

Stronger BERT fine-tune

Hyperparameter sweep + 5 epochs to close the gap to published ~0.85 F1; learning-rate warmup, layer-wise LR decay.

Error analysis on confusable pairs

Targeted study of `ibm.pc` vs `mac.hardware` and religion clusters — possibly with contrastive fine-tuning.

Larger Llama with few-shot $k > 1$

Test 70B model, retrieve nearest-neighbour demonstrations per query rather than fixed examples.

Thank you

Questions are welcome.

Part 1 · Simple NN with TF-IDF / W2V / GloVe

Part 2 · Fine-tuning BERT (full / probe / partial)

Part 3 · Domain-adaptive MLM then fine-tune

Part 4 · Llama-3 zero-shot and few-shot prompting